Jindong Cao

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Education

Carnegie Mellon University, College of Engineering Pittsburgh, USA
M.S. in Electrical and Computer Engineering, GPA: 3.75/4.0 *Aug 2023 – Dec 2024*
Sun Yat-sen University, School of Computer Science and Engineering Guangzhou, China
B.S. in Computer Science and Technology, GPA: 3.9/4.0 *Aug 2019 – Jul 2023*

Experience

TikTok San Jose, USA
Machine Learning Engineer *Jan 2025 – Present*

- Received **Exceeds Expectations Performance Rating** (Top 5% performance rating) and **fast promotion**.
- Led migration of core push systems to a unified C++ scheduling framework, decoupling business logic from global decision-making and delivering 320,000 DAU.
- Integrated US e-commerce interest candidates into real-time notification pipelines, delivering 400 million USD GMV.
- Owned end-to-end push modeling pipeline, including feature engineering, sample/label construction, model design, and online integration with ranking and frequency control.
- Upgraded ranking model from single-target CTR to multi-task learning (Shared-Bottom + Multi-Tower), integrating post-click signals (Like, Finish) to reduce clickbait and improve DAU.

Amazon San Diego, USA
Software Development Engineer Intern *May 2024 – Aug 2024*

- Built internal full-stack tools with React, Java, and TypeScript, improving service accessibility and operational efficiency.

ZMO.AI Shenzhen, China
Machine Learning Engineer Intern *Mar 2023 – May 2023*

- Developed an AI image editing pipeline in Python integrating ChatGPT, SAM, GroundingDINO, and Stable Diffusion.
- Applied pruning to BLIP-2 and CLIP, reducing memory usage by 30% and runtime by 40%.

Huawei Technologies Dongguan, China
Software Development Engineer Intern *Jul 2022 – Sep 2022*

- Refactored a data center query system in C++, reducing average query latency by 70%.

Research

Pattern Recognition (SCI Q1) Guangzhou, China
Co-author, weakly supervised prompt learning for low-resource medical image classification *Jan 2024*

- Led experiment design and execution, and contributed to manuscript writing.
- Leveraged transferable representations from large-scale pre-trained vision-language models to address low-resource medical image classification for clinical decision support.
- Improved zero-shot and few-shot classification performance via additional prompt-embedding structures.

Awards & Skills

Challenge Cup Competition of Science and Technology Guangzhou, China May 2022

- National First Prize (ranked 130 / 300,000).
- Project focus: object detection for autonomous robots using black-and-white imagery.

Skills: Python, C++, SQL, PyTorch, TensorFlow, Deep Learning, Machine Learning, Recommendation System